Band-1 Andrews-Curtis Presentation Probe (Collapse 2.33)

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Band-1 Andrews-Curtis – Toy Trivializations + Named Open AK(3)

Honesty stamp. This note does **not** claim the Andrews-Curtis conjecture is solved. It does **not** claim AK(3) is AC-trivialized. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope. Toy AC-trivializations on box are certificates for **tiny** balanced presentations of the trivial group only.

> **Locked CERTIFIED (box):** $\langle a, b \mid ab,\, b\rangle$ AC-trivializes in **1** move; $\langle a, b \mid ab,\, a\rangle$ AC-trivializes in **2** moves (BFS).
> **Locked OPEN (named):** AK(3) = $\langle a, b \mid a^3b^{-4},\, abab^{-1}a^{-1}b^{-1}\rangle$ – ordinary AC unresolved; bounded box BFS **not** a counterexample.
> **Full AC conjecture: OPEN**.

Abstract

Andrews-Curtis (1965) asks whether every balanced presentation of the trivial group is related to the standard presentation $\langle x_1, \ldots, x_n \mid x_1, \ldots, x_n\rangle$ by a finite sequence of AC-moves: (AC1) multiply one relator by another; (AC2) invert a relator; (AC3) conjugate a relator by a word. Potential counterexamples of Akbulut-Kirby type remain the frontier; **AK(3)** (total relator length 13) is the classical named open case for ordinary AC.

This Band-1 probe certifies two hand/BFS toy trivializations and ledgers AK(3) as **OPEN**. Shallow non-finding on AK(2)/AK(3) is **not** a counterexample certificate.

Problem (collapse Band-2 item 2.33 → Band-1 finite + lit)

1. State AC moves and the balanced-trivial-group conjecture.
2. Certify one or two tiny presentations that AC-trivialize on box.
3. Name **AK(3)** as the remaining open class for this probe.
4. Keep Clay / ToE / GATE-04 / "AC solved" off the claim surface.

Canonical statement

A presentation $\langle x_1, \ldots, x_n \mid r_1, \ldots, r_n\rangle$ is **balanced**. It presents the **trivial group** if the normal closure of $\{r_i\}$ is the free group F_n. **AC-moves** on the relator tuple:

Move	Operation
AC1	$r_i := r_i \cdot r_j$ $(i \neq j)$
AC2	$r_i := r_i^{-1}$
AC3	$r_i := w^{-1} r_i w$ (word w in the generators)

Andrews-Curtis conjecture: every balanced presentation of the trivial group can be reduced to $\langle x_1, \ldots, x_n \mid x_1, \ldots, x_n \rangle$ by finitely many AC-moves.

Literature ledger (web check; not re-proved on box)

Year / item	Result	Status
1965	Andrews-Curtis conjecture stated	classical
1985	Akbulut-Kirby potential counterexamples AK(n)	Topology **24**
–	AK(2) known AC-trivializable	lit (not a counterexample)
–	Rank-2 verified through total relator length **12**; length **13** reduces to AK(3) class	computational lit
2024	Shehper et al. arXiv:2408.15332 (stable-AC / AI search context)	context – **not** elevated to ordinary-AC closure
2026	Carreras arXiv:2607.23611 length-14 certificates; several connect to AK(3)	**CLAIMS / partial** – AK(3) still open
–	Ordinary AC for **AK(3)**	**OPEN**

Hygiene. Stable-AC literature is nuanced and contested; this probe tracks **ordinary** AC only. Do not elevate preprint "solved" language.

Method (box)

1. Represent words as reduced integer tuples over generators $\{\pm 1, \pm 2\}$ for $\{a^{\pm 1}, b^{\pm 1}\}$.
2. BFS over AC1 / AC2 / AC3 (box AC3 restricted to conjugators of generator length 1) until a presentation with relators equal (up to inverse) to the free generators is reached, or visit/bound caps hit.
3. Hand sequences for the two toys cross-check BFS.
4. Bounded BFS on AK(2) and AK(3): record non-finding with visited/wall – **explicitly not** a counterexample.

Lab: `mission-control/experiments/band1/andrews_curtis/` (`probe_ac.py`, `results.json`).

Box results

Toy 1 – $\langle a, b \mid ab,\, b \rangle$ **(trivial group)**

Item	Value
Hand	AC1: $r_1 * = r_2^{-1} \to \langle a, b \mid a,\, b \rangle$
BFS	**TRIVIALIZED** in **1** move (`AC1:r1*=r2^{-1}`); visited 3; wall ~0.0002 s
Status	**AC-TRIVIALIZED (certified)**

Toy 2 – $\langle a, b \mid ab,\, a \rangle$ **(trivial group)**

Item	Value
Hand	AC2 invert r_1; AC1 $r_1* = r_2 \to \langle a, b \mid b^{-1}, a\rangle$
BFS	**TRIVIALIZED** in **2** moves; visited 23; wall ~0.0014 s
Status	**AC-TRIVIALIZED (certified)**

AK(2) shallow (known trivializable in lit)

Presentation $\langle a, b \mid aab^{-1}b^{-1}b^{-1}, abab^{-1}a^{-1}b^{-1}\rangle$. Box BFS: **NOT_FOUND_IN_BOUNDS** (visited 40001; wall ~2.36 s). **Non-finding != counterexample**; lit already marks AK(2) AC-trivializable.

Named open – AK(3)

$$\langle a, b \mid a^3b^{-4}, abab^{-1}a^{-1}b^{-1}\rangle$$

Item	Value
Total relator length	13
Box BFS	**NOT_FOUND_IN_BOUNDS** (visited 40001; wall ~2.81 s)
Status	**OPEN** – ordinary AC unresolved
Claim	Ledger only; **not** a counterexample certificate

What is proved vs open (statement lock)

Certified here

1. Two tiny balanced trivial presentations AC-trivialize under the box move set (hand + BFS).
2. Probe infrastructure (normalize, BFS, honesty flags) is reproducible from `results.json`.

Locked OPEN

1. **Andrews-Curtis conjecture** in full – **OPEN**.
2. **AK(3)** ordinary AC-trivialization – **OPEN** (named remaining class for this probe).
3. Length-14 / MS hard cases connecting to AK(3) – open frontier per lit ledger.

Explicit non-claims

- AC conjecture **not** claimed solved.
- AK(3) **not** claimed trivialized or refuted.
- No Clay; no ToE; **GATE-04 LOCKED**.
- Bounded BFS non-finding is **not** a counterexample.

Why Band-1 vs full conjecture

- **Band-1 (this probe):** honest finite certificates on toys + named open AK(3).
- **Band 3-4:** unrestricted AC / potential counterexample resolution.
- Collapse note: "many tiny cases easy; AK(3) class still open" – reflected honestly.

Deliverables

- Lab: `mission-control/experiments/band1/andrews_curtis/` (`probe_ac.py`, `results.json`).
- Paper/PDF: `papers/2026-09-11_Band1_Andrews_Curtis_presentation_probe.{md,pdf}` → `DOWNLOAD_PDFs/` + Mac `Downloads/DOWNLOAD_PDFs_2026-09-11/`.
- House `next_wave.md` updated.

References (primary)

1. J. J. Andrews, M. L. Curtis, *Free groups and handlebodies*, Proc. Amer. Math. Soc. **16** (1965).
2. S. Akbulut, R. Kirby, *A potential smooth counterexample in dimension 4 to the Poincaré conjecture, the Schoenflies conjecture, and the Andrews-Curtis conjecture*, Topology **24** (1985), 375-390.
3. M. Shehper et al., arXiv:2408.15332 (stable-AC / search context – not elevated here).
4. A. Carreras, arXiv:2607.23611 (length-14 certificates; AK(3) still open – CLAIMS/partial).